

Tools for software development

Lab 3 - Test Design and Test Management



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**PLAN
DESIGN
REPORT**



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Where we are in the course

The project moves from setup to professional test documentation.



Lab 3 outcomes

By the end of the lab, each team should have concrete test artifacts.

- Understand what a test plan is and why it is useful
- Create a reusable test case template
- Apply black-box and experience-based test design techniques
- Design 10–15 manual test cases for the semester project
- Write at least one defect report
- Select 2–3 test cases suitable for Selenium automation

Main output

- Test plan
- Test case template
- Manual test cases
- Defect report
- Automation candidates



Why Lab 3 matters

Automation without good manual test design usually produces fragile tests.

Without Lab 3

- unclear scope
- random testing
- weak automation candidates
- unclear expected results
- hard final defense

With Lab 3

- defined test scope
- repeatable test cases
- clear priorities
- documented defects
- ready for Selenium/Robot/Postman



Test plan: minimum viable version

A short plan is enough if it helps the team test systematically.

1

Tested application

name, URL, brief purpose

2

Scope

what is tested / not tested

3

Objectives

what the team wants to verify

4

Test types

manual, UI, API, regression

5

Environment

browser, OS, tools

6

Risks

instability, missing API, dynamic UI



Test case anatomy

Every test case should be executable by someone else.

Field	Purpose
ID	Unique identifier, e.g., TC-001
Title	Short descriptive name
Preconditions	What must be true before execution
Test data	Inputs, user accounts, IDs, products
Steps	Numbered actions to perform
Expected result	Observable expected behavior
Actual result + status	Filled during execution

TC-001 - Login with valid credentials

Precondition: user is on login page

Data: standard_user / secret_sauce

Steps:

1. Enter username

2. Enter password

3. Click Login

Expected: product page is displayed

Rule: if another team member cannot execute it, the test case is not written well enough.



Test design techniques

Use multiple techniques, not only intuition.

Technique	Use it for	Example
Boundary value	limits and edges	60/61 points, min/max length
Equivalence partitioning	groups of valid/invalid inputs	valid email vs invalid email
Truth table	combinations of conditions	role + login + account state
State transition	stateful flows	cart: empty → filled → checked out
Use case testing	real user workflows	search → detail → add to cart
Exploratory testing	learning the product	find unexpected behavior



Boundary value + equivalence partitioning

Small input changes often reveal defects.

Example: username length

- Requirement: username must have 3–20 characters.
- Equivalence classes:
 - invalid: 0–2 characters
 - valid: 3–20 characters
 - invalid: 21+ characters
- Boundary values:
- 2, 3, 20, 21

Example test cases

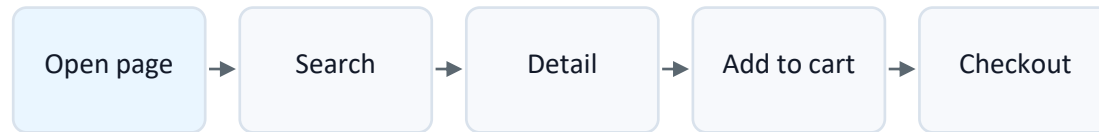
- TC-001: username with 2 chars → error
- TC-002: username with 3 chars → accepted
- TC-003: username with 20 chars → accepted
- TC-004: username with 21 chars → error



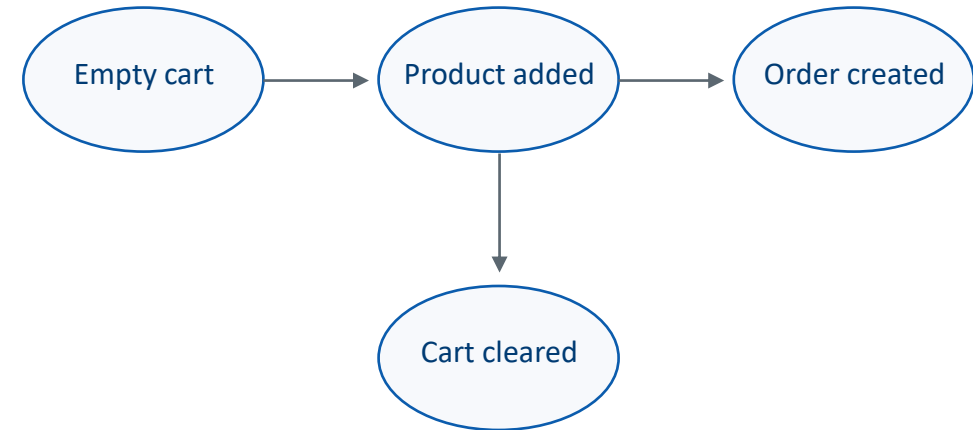
Use case and state transition tests

Good for flows such as login, cart, booking, or order processing.

Use case flow



State transition



Ask: what are the important paths and what invalid transitions should be blocked?



Exploratory testing

A structured way to learn the product and discover defects.



- Focus on a risk or area of the system
- Take notes while testing
- Record questions and unexpected behavior
- Finish with a short session report
- Convert useful findings into formal test cases



Exercise 1: create a test case template

Teams define the format they will use for all manual tests.

```
# Test Case Template

| Field | Description |
|---|---|
| Test Case ID | TC-001 |
| Title | Short test name |
| Feature | Tested functionality |
| Priority | High / Medium / Low |
| Type | Functional / UI / Negative |
| Preconditions | Before execution |
| Test Data | Inputs used |
| Steps | Step-by-step actions |
| Expected Result | What should happen |
| Actual Result | Filled during execution |
| Status | Not Run / Passed / Failed |
```

File to create

manual-tests/test-case-template.md

Tip

Keep the template simple. The goal is repeatability, not paperwork.



Patterns for preparing templates

Use for design own...

- Google: "test cases" xls "
- Example of an xls file:
 - <https://www.guru99.com/download-sample-test-case-template-with-explanation-of-important-fields.html>

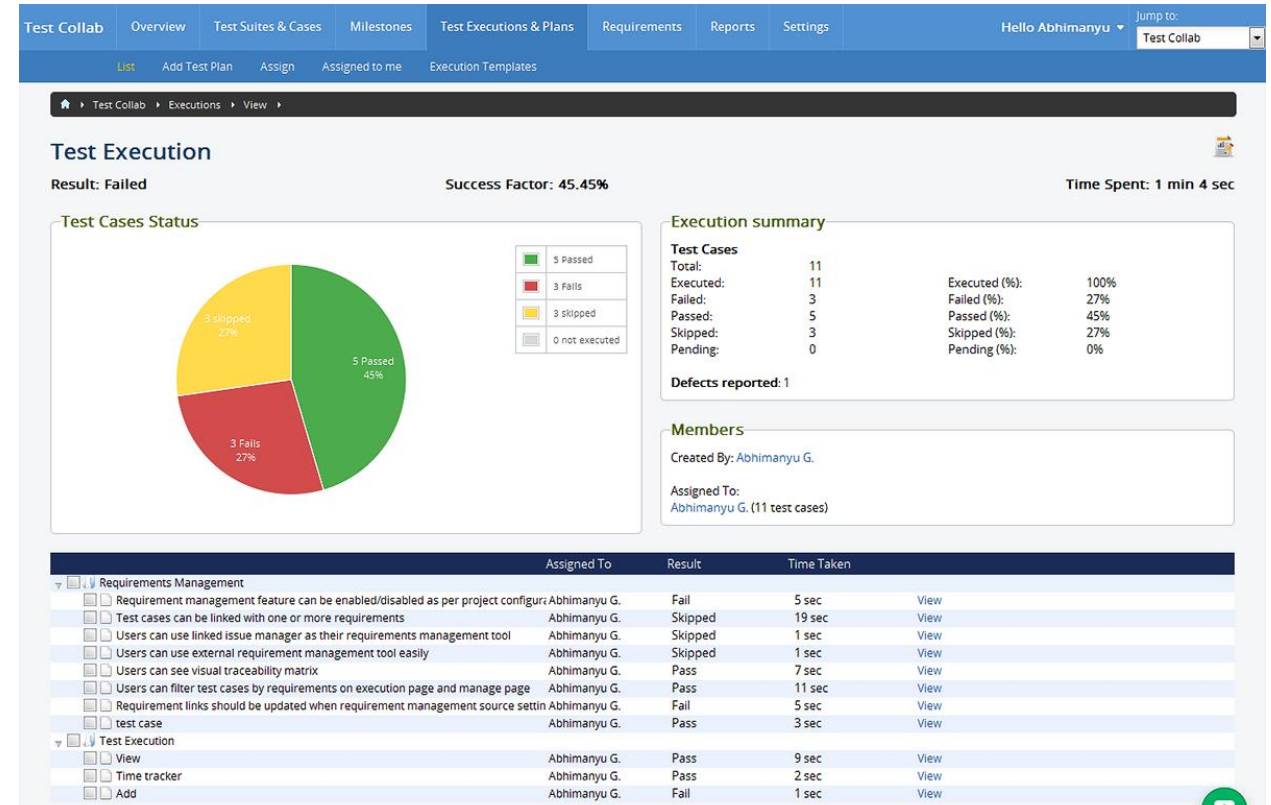
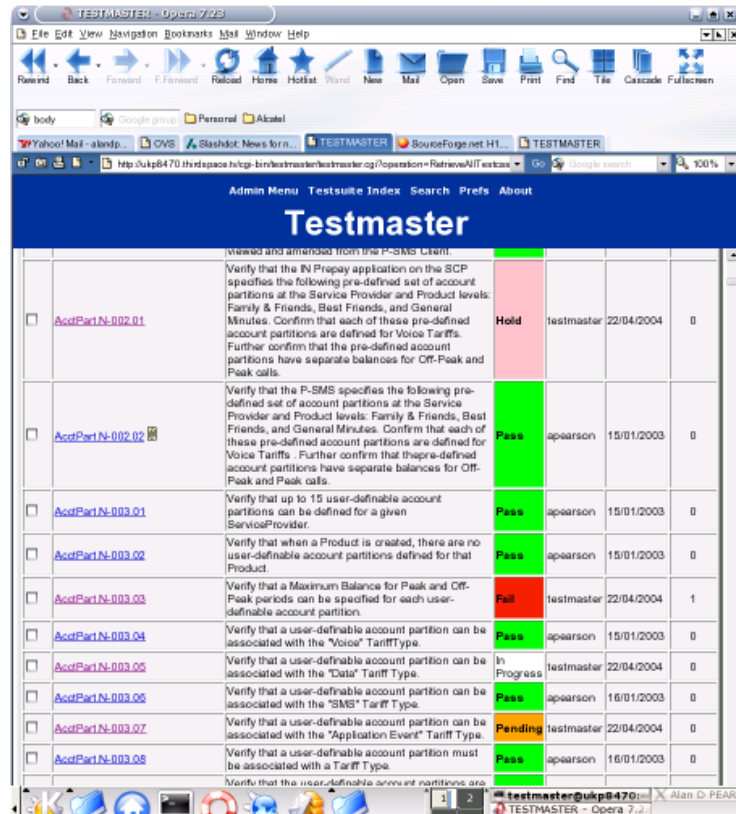
	A	B	C	D	E	F	G	H	I	J	K
1	Test Case ID		BU_001	Test Case Description		Test the Login Functionality in Banking					
2	Created By		Mark	Reviewed By		Bill	Version		2.1		
3											
4	QA Tester's Log		Review comments from Bill incorporate in version 2.1								
5											
6	Tester's Name		Mark	Date Tested		1-Jan-2017		Test Case (Pass/Fail/Not		Pass	
7											
8	S #	Prerequisites:				S #	Test Data				
9	1	Access to Chrome Browser				1	Userid = mg12345				
10	2					2	Pass = df12@434c				
11	3					3					
12	4					4					
13											
14	Test Scenario		Verify on entering valid userid and password, the customer can login								
15											
16	Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended		
17											
18	1	Navigate to http://demo.guru99.com		Site should open		As Expected			Pass		
19	2	Enter Userid & Password		Credential can be entered		As Expected			Pass		
20	3	Click Submit		Cutomer is logged in		As Expected			Pass		
21	4										
22											
23											
24											
25											
26											
27											



Test case recording systems

- desktop example: <http://testmaster.sourceforge.net/>
- web example: <https://testcollab.com/>

Test Collab



Exercise 2: design manual test cases

Create a first complete test suite for your selected project.

Category	Minimum
Positive scenarios	3
Negative scenarios	3
Boundary / equivalence tests	2
Use case / workflow tests	2
Total	10–15 test cases

Required file

- manual-tests/manual-test-cases.md
- Use at least three test design techniques.
- Mark test cases that could later be automated.

Quality over quantity: 10 good tests are better than 25 unclear tests.



Example: e-shop test cases

Useful for projects such as Sauce Demo, Demo Web Shop, or OpenCart.

ID	Test case	Technique
TC-001	Login with valid credentials	Use case
TC-002	Login with invalid password	Negative
TC-003	Login with empty username	Validation
TC-004	Search product by keyword	Use case
TC-005	Search product with no results	Equivalence
TC-006	Add product to cart	State transition
TC-007	Remove product from cart	State transition
TC-008	Checkout with empty required fields	Boundary / validation



Example: API test cases

Useful for projects such as JSONPlaceholder, Reqres, DummyJSON, or Restful Booker.

ID	Test case	Technique
TC-001	GET list of users returns 200	Positive API test
TC-002	GET existing user by ID	Equivalence
TC-003	GET non-existing user ID	Negative
TC-004	POST create user with valid data	Use case
TC-005	POST create user with missing field	Validation
TC-006	PUT update user	CRUD workflow
TC-007	DELETE user	State transition
TC-008	Verify response schema	Structural validation



Priorities and risk

Not all test cases are equally important.

High priority

- Core user journey
- Login/authentication
- Checkout/order/reservation
- Data creation/update
- Known risky areas

Medium priority

- Common alternatives
- Filtering and sorting
- UI messages
- Non-critical validations
- Browser differences

Low priority

- Rare flows
- Cosmetic checks
- Optional features
- Nice-to-have scenarios
- Future extensions

Use priority to decide what to automate first in Lab 4.



Defect report

A defect report must help someone reproduce and understand the issue.

Field	Example
Defect ID	BUG-001
Title	Checkout allows empty last name
Related test	TC-009
Severity / Priority	Major / High
Environment	Chrome, Windows 11
Steps to reproduce	numbered and precise
Expected vs actual	what should happen vs what happened
Evidence	screenshot, log, URL

Good report

A developer can reproduce the bug without asking additional questions.

Bad report

“It does not work.”



Exercise 3: write a defect report

Find a real issue or create a realistic example based on a failed test case.

```
# BUG-001 - Checkout allows empty last name
```

```
Related Test Case: TC-009
```

```
Severity: Major
```

```
Priority: High
```

```
Environment: Chrome, Windows 11
```

```
Steps to reproduce:
```

1. Add product to cart
2. Open checkout
3. Fill first name and postal code
4. Leave last name empty
5. Click Continue

```
Expected: validation error is displayed
```

```
Actual: user continues to the next step
```

Required file

reports/defect-report-example.md

Evidence

Attach screenshot or provide exact URL/log if possible.



Team workflow in Git

Keep documentation structured and reviewable.

```
tsd-2026-team-name/  
├── docs/  
│   ├── project-selection.md  
│   ├── test-plan.md  
│   └── test-plan.xlsx  
├── manual-tests/  
│   ├── test-case-template.md  
│   └── manual-test-cases.md  
├── reports/  
│   └── defect-report-example.md  
└── automation/  
    ├── selenium/  
    ├── robot/  
    └── postman/
```

- Work in branches when editing shared documents
- Use meaningful commit messages
- Review at least one test case from another team member
- Do not overwrite project-selection.md from Lab 2
- Prepare test cases for automation in Lab 4



Evaluation for Lab 3

Lab 3 is the main documentation checkpoint.

Test plan .md - Test case template .xlsx

5 point

10–15 meaningful test cases

4 point

Use of design techniques,

3 point

Defect report your app (filled test plan (xlsx) by running test cases)

2 point

Repository/documentation quality

1 point

Total: 15 points

Deadline: end of Lab 3 / end of day 6.7.2026.



Prepare for Lab 4: Selenium

Select test cases that are good candidates for UI automation.

- Select 2–3 manual test cases for automation
- Prefer stable, repeatable workflows
- Prepare test data and accounts
- Identify important UI elements and expected assertions
- Mark tests that should not be automated and explain why

Good candidates

- Login
- Search
- Add to cart
- Form validation
- Navigation between pages



Lab 3 wrap-up

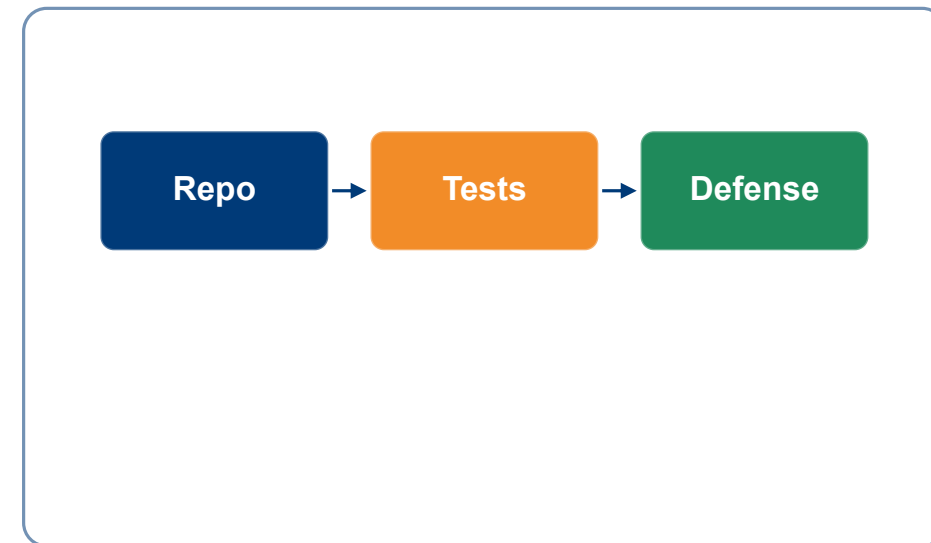
Your test cases are the input for the rest of the course.

- Test plan defines the project scope
- Manual test cases become automation candidates
- Defect reports support final project documentation
- Priority helps decide what to automate first
- Next lab: Selenium UI automation

Next lab

Lab 4 – **test design techniques and manual test cases**





Thank you for your attention.

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